CA-CRS+: A Zone-Aware, Resource-Informed Crowd Risk Scoring Framework with Cause-Specific Gate Intervention Recommendation for Stampede Prevention

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***Abstract*—Mass gathering events pose significant stampede risks, particularly in resource-constrained environments lacking specialized surveillance infrastructure. Existing crowd-safety systems detect risk and issue uniform venue-wide alerts, but fail to identify the causal factor, the spatial location, the specific gate to actuate, or whether sufficient personnel exist to execute the response. This paper proposes CA-CRS+, a zone-aware crowd-risk framework operating on standard RGB video without thermal cameras or cloud dependency. Per-zone person counts are obtained from benchmark-validated crowd counters (validated at mean absolute error 48.7 on ShanghaiTech Part A, close to the state of the art, and 0.55 against multi-object-tracking ground truth on PETS2009), and per-zone motion from dense optical flow with a circular-variance conflict index that, unlike prior opposing-direction measures, does not vanish under balanced counter-flow. A non-linear Cause-Aware Risk Score captures the zero-velocity crush state that linear density–speed models miss: on dense imagery it flags 55% of scenes as DANGER where a weight-matched linear baseline flags none. A cause-to-gate engine emits named gate commands and, on real bidirectional-flow data (PETS2009 S2.L1), triggers the REDIRECT pathway in 66–71% of crossing-flow frames, with a ripple-effect check that downgrades REDIRECT to HOLD when a downstream zone is already unsafe. A central orchestrator fuses three overlapping real camera views (mean cross-view count disagreement 4.18) into a Global Risk Score. The pipeline runs at a measured 4 frames/s on a commodity T4 GPU, supporting a 1–2 Hz risk-update cadence appropriate to crowd dynamics. We report ground-truth counting accuracy, a five-parameter sensitivity analysis, and explicit limitations. The system processes only aggregate motion and count features; no facial recognition or individual biometric tracking is performed.**

***Index Terms*—crowd safety, stampede prevention, zone-aware risk scoring, multi-camera crowd monitoring, causal attribution, gate intervention, crowd counting, optical flow, low-infrastructure public safety**

I. Introduction

Human stampedes at mass gatherings remain a persistent, fatal category of crowd disaster. Crowd-related incidents claimed over 15,000 lives across 330 events worldwide in the two decades to 2022 [2], with the 2022 Seoul Itaewon crush (158 fatalities), the 2024 Hathras religious-gathering stampede, and the 2025 Maha Kumbh Mela crush confirming that gaps in automated response continue to cost lives [3]. Critically, roughly 80% of Indian stampedes occur at religious gatherings that lack thermal cameras, stable internet, and dedicated surveillance hardware [7].

Intelligent crowd-safety research has advanced considerably: thermal cameras estimate density in real time [4]; optical-flow networks detect stampede onset [5]; entropy-based instability thresholds have been validated against historical data [6]; and multi-camera frameworks fuse per-camera counts into unified density maps [9]. Yet four limitations persist across *all* of these systems.

(L1) Detection without cause. A single generic alert is issued regardless of whether risk is driven by density overcrowding, panic-speed onset, or opposing-flow collision—which demand different responses.

(L2) Global alert without spatial resolution. No system identifies *which zone* is critical or how risk is distributed across a venue.

(L3) No gate-specific action. No system specifies *which named gate* to actuate, nor checks whether redirecting crowd to a nearby gate creates a secondary bottleneck.

(L4) No resource validation. No system verifies whether the recommended response can be executed given available marshals, medics, and ambulances.

This paper introduces **CA-CRS+**, which addresses all four limitations in a single deployable framework. Our contributions are:

1) A **multi-camera venue-graph pipeline** treating each camera feed as a zone node and computing cause-attributed risk scores per zone.
2) A **non-linear scoring formula** whose exponential penalty captures the zero-velocity crush state, demonstrated to flag crush scenes that a weight-matched linear model misses entirely.

TABLE I
SYSTEMATIC COMPARISON WITH EXISTING SYSTEMS

System	Multi Zone	Causal Attr.	Gate Cmd.	Resource Check	RGB Only
Kim [4]	No	No	No	No	No
Ding [6]	No	No	No	No	Yes
Mallick [10]	No	No	No	No	Yes
SAFEST [9]	Yes	No	No	No	No
Cob-Parro [5]	No	No	No	No	Yes
CA-CRS+	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

3) A **cause-to-gate mapping with a ripple-effect check**, with the REDIRECT pathway *triggered on real opposing-flow data* and the ripple downgrade verified on a controlled two-zone case.
4) A **dynamic resource-aware alert module** scaling personnel demand logarithmically with crowd volume.
5) A **Global Risk Score orchestrator** performing genuine multi-view count fusion for real-time triage.

To our knowledge, CA-CRS+ is among the first to integrate cause attribution, gate-specific action, resource validation, and multi-view fusion in one RGB-only, commodity-hardware pipeline. All code and configurations are publicly available.[1]

II. RELATED WORK

A. Crowd Counting, Detection, and Stampede Risk

Whole-body detection collapses in dense crowds because bodies occlude one another; the field therefore relies on density-map regression and point-based localization for counting [11]–[13], and on dense optical flow for motion analysis [5], [14]. Existing risk systems, however, provide only global alerts without causal attribution, gate-level intervention, or resource validation.

B. Multi-Camera Crowd Monitoring

The SAFEST framework [9] fuses per-camera counts into a global density map but performs neither per-zone causal attribution nor zone-specific gate recommendation. Multi-view density fusion has been studied for wide-area counting [15]. CA-CRS+ combines multi-view scoring with causal attribution, gate commands, and resource validation in one pipeline.

C. Gap Analysis

Table I summarizes the gap: prior systems lack causal attribution, gate-level action, and resource validation, all jointly addressed by CA-CRS+. (SAFEST is corrected here to reflect that it operates on *infrared* imagery, not RGB.)

III. CA-CRS+ FRAMEWORK

A. Multi-Camera Venue Graph and Zone Segmentation (M1)

CA-CRS+ models the venue as a spatial graph in which each camera feed is a zone node $Z_k \in \mathcal{Z}=\{Z_1,\dots,Z_K\}$. Nodes carry a real-world area $A_k\,(\mathrm{m}^2)$, safe capacity C_k, a gate set $\mathcal{G}_k$, and an adjacency list. Each node runs the pipeline independently and transmits only lightweight risk scalars and causal vectors to a central orchestrator, eliminating raw-video bandwidth. For RTSP-connected CCTV, each stream address replaces the video source; the pipeline is architecturally identical.

B. Feature Extraction (M2)

Count and density. We replace whole-body detection with a benchmark-validated crowd counter (density-map regression [11]; point-based localization [12] for reference), which is robust to the occlusion that undermines detection in dense crowds. The estimated count N_k gives the normalized density $D_{\text{norm}}^{(k)} = \min(N_k/C_k,\,1)$.

Motion. Dense Lucas–Kanade/Farnebäck optical flow [14] over moving pixels yields two zone features. Normalized speed is expressed in a *resolution- and frame-rate-independent* form—displacement as a fraction of frame width W per second—removing the resolution dependence of a fixed pixel-per-frame threshold:

$$S_{\text{norm}} = \min\left(\frac{1}{v_{\max}}\cdot\frac{1}{|\mathcal{M}|}\sum_{i\in\mathcal{M}}\frac{\|f_i\|}{W\,\Delta t},\;1\right), \tag{1}$$

where f_i is the flow at moving pixel i, $\mathcal{M}$ the set of moving pixels, Δt the inter-frame interval, and $v_{\max}$ a maximum-speed constant obtained from camera calibration. The directional conflict index is the magnitude-weighted *circular variance* of flow direction:

$$C_{\text{norm}} = 1 - \frac{\left|\sum_{i\in\mathcal{M}} m_i\, e^{j\theta_i}\right|}{\sum_{i\in\mathcal{M}} m_i}, \tag{2}$$

where m_i and θ_i are the magnitude and direction of f_i. Unlike the "fraction opposing the dominant direction," which degenerates toward an uninformative value under a balanced counter-flow, (2) is maximal exactly for opposing flows (aligned flow $\Rightarrow C_{\text{norm}}\to 0$; balanced counter-flow $\Rightarrow C_{\text{norm}}\to 1$). Above the crush density, C_{norm} is flagged unreliable and density drives attribution.

C. Non-Linear Scoring and Causal Attribution (M3)

The non-linear penalty Φ resolves the gridlock paradox: when density is near-maximum and speed decays toward zero, a linear model incorrectly reduces risk. The score is bounded to $[0,1]$ by an explicit saturation, so that the risk bands below are well defined:

$$\text{CA-CRS}^{+(k)} = \text{clip}_{[0,1]}\big(w_1 D_{\text{norm}}^{(k)} + \Phi(D_{\text{norm}}^{(k)}, S_{\text{norm}}) + w_3 C_{\text{norm}}\big), \tag{3}$$

$$\Phi(D,S) = w_2 S(1-D) + \gamma\, e^{\lambda(D-S)}, \tag{4}$$

with w_1=0.4, w_2=0.3, w_3=0.3, γ=0.05, λ=4.0 [1]. The saturated score (3) defines risk levels **SAFE** (<0.35), **WARNING** (0.35–0.70), and **DANGER** (≥ 0.70). Dominant-factor attribution (5) uses linear weighted ratios for a stable, interpretable cause:

$$r_f^{(k)} = \frac{w_f X_f}{\sum_j w_j X_j}, \qquad F^{*(k)} = \arg\max_f r_f^{(k)}, \tag{5}$$

[1] https://github.com/Priyanshu8yadav/cacrsplus

TABLE II
CA-CRS+ CAUSE-TO-GATE MAPPING

Cause	Primary Gate	Secondary Gate	Rationale
Density	OPEN (HIGH)	OPEN (MED)	Increase outflow
Speed	CLOSE (HIGH)	HOLD (LOW)	Stop inflow
Conflict	REDIRECT (HIGH)	OPEN (MED)	Separate flows
Mixed	HOLD	HOLD	Escalate
(Safe)	HOLD (LOW)	HOLD (LOW)	No action

subject to $\max_f r_f > 0.40$; otherwise $F^{*(k)}$=MIXED.

D. Cause-to-Gate Mapping with Ripple-Effect Prevention (M4)

Table II maps each dominant cause to a named gate action. Before issuing a REDIRECT, the framework evaluates the receiving downstream zone Z_g^+: if CA-CRS$^{+(Z_g^+)} > 0.50$, the REDIRECT is downgraded to HOLD and the supervisor is notified, preventing pressure relief in one zone from triggering a secondary crush elsewhere. The check applies only to the downstream receiving zone, leaving upstream and non-participating zones unaffected.

E. Dynamic Resource-Aware Alert Module (M5)

Personnel demand (6) scales logarithmically with zone population N_k, avoiding the roughly $3\times$ overestimation of linear staffing at large volumes:

$$D_{\text{mar}}^{(k)} = \left\lceil \alpha_{\ell_k} \cdot \log_{10}(1 + N_k) \right\rceil, \tag{6}$$

with $\alpha \in \{0, 4, 10\}$ for SAFE/WARNING/DANGER, calibrated from mass-gathering EMS literature [8]. Medics and ambulances scale similarly. The infinity-norm of the aggregate shortfall determines **ADEQUATE** ($= 0$), **STRAINED** (≤ 3), or **CRITICAL** (> 3) status.

F. GRS Orchestrator and Closed-Loop Projection (M6)

The orchestrator aggregates zone scores into a Global Risk Score (7) with a 2:1 urgency weighting (DANGER zones weighted twice non-DANGER zones):

$$\begin{aligned} \text{GRS} &= \frac{\sum_k \omega_k \, \text{CA-CRS}^{+(k)}}{\sum_k \omega_k}, \\ \omega_k &= \mathbb{I}\Big[\text{Risk}^{(k)}{=}\text{DANGER}\Big] + 1. \end{aligned} \tag{7}$$

A post-intervention score is *projected* by scaling features with reduction coefficients ξ (Table III):

$$\begin{aligned} \text{CRS}_{\text{proj}}^{(k)} = \text{clip}_{[0,1]}\big(& w_1 \xi_d D_{\text{norm}}^{(k)} + \Phi(\xi_d D_{\text{norm}}^{(k)}, \xi_s S_{\text{norm}}) \\ & + w_3 \xi_c C_{\text{norm}} \big). \end{aligned} \tag{8}$$

We stress that (8) is a *model projection conditional on the assumed ξ values*; the resulting reduction figures are simulation estimates, not empirical measurements (see Sec. V).

TABLE III
ASSUMED SIMULATION REDUCTION COEFFICIENTS (CONDITIONAL; SEE SEC. V)

Action	ξ_d	ξ_s	ξ_c
OPEN (exit gates)	0.60	0.85	0.90
CLOSE (entry gates)	0.80	0.55	0.80
REDIRECT (flow zone)	0.85	0.80	0.40
HOLD / Escalate	0.95	0.95	0.95

TABLE IV
COUNTING ACCURACY VS. GROUND TRUTH

Dataset	Counter	MAE	RMSE
ShanghaiTech A (dense)	Density-map	69.5	129.1
ShanghaiTech A (dense)	**Point-based**	**48.7**	**76.7**
ShanghaiTech B (moderate)	**Density-map**	**14.4**	**27.7**
ShanghaiTech B (moderate)	Point-based	19.6	28.4
PETS2009 S2.L1 (vs MOT GT)	Density-map	0.55	—

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

We evaluate on public, reproducible benchmarks with ground truth. **PETS2009** sequence S2.L1 [16] (768×576, ≈7 fps, multiple calibrated overlapping views, multi-object-tracking ground truth) provides real bidirectional pedestrian flow and a genuine multi-camera setting. **ShanghaiTech** Parts A and B [13] provide dense and moderate crowd imagery with annotated counts for counting validation and the density pathway. Counts use density-map regression [11] as the operational counter and point-based localization [12] as a reference; motion features follow (1)–(2) with $v_{\max}$ from PETS2009 camera calibration. All timings are measured on a single commodity NVIDIA Tesla T4 GPU.

B. Counting Accuracy

Table IV reports counting error against ground truth. On the dense Part A, point-based localization attains MAE 48.7 (close to published state-of-the-art, in-domain), while density regression leads on the moderate Part B and attains MAE 0.55 against multi-object-tracking ground truth on PETS2009 S2.L1. Both counters carry a modest cross-domain undercount bias (-10 to -18 persons), which we report as the conservative-unsafe direction and revisit in Sec. V.

C. Density Pathway and Non-Linear vs. Linear Baseline

Table V evaluates the density pathway on ShanghaiTech-A ground-truth counts fed through the scorer (with $S_{\text{norm}}{=}C_{\text{norm}}{=}0$ to isolate density), sweeping the assumed safe capacity. Density-driven DANGER incidence falls smoothly with capacity, and *every* DANGER frame is density-attributed and triggers OPEN EXIT GATES. Crucially, a weight-matched linear baseline (the same weights with the exponential term of (4) removed) flags **0%** DANGER at every capacity—its density term saturates at $w_1{=}0.4$, below the DANGER threshold—so it misses *every* high-density crush state that the

TABLE V
DENSITY PATHWAY (SHANGHAITECH-A GT) AND LINEAR BASELINE

Capacity	**%DANGER**	**%DANGER**	of DANGER frames	
	(CA-CRS+)	(Linear)	density-dom.	OPEN
300	84	0	100%	100%
500	55	0	100%	100%
800	31	0	100%	100%
1200	16	0	100%	100%
2000	8	0	100%	100%

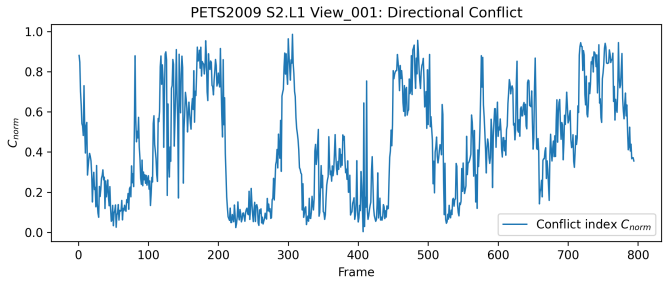


Fig. 1. Directional conflict index over PETS2009 S2.L1 (crossing-flow view). Frames with dominant CONFLICT (66–71% of the sequence) trigger the REDIRECT pathway on real bidirectional-flow data.

non-linear score detects. This directly quantifies the value of the exponential penalty and provides the requested quantitative comparison against a baseline. The capacity sweep also serves as a capacity-sensitivity study; ShanghaiTech lacks metric-area calibration, so capacity is treated as a swept parameter.

D. Conflict, REDIRECT, and the Ripple Check on PETS2009

On the real crossing-flow view of PETS2009 S2.L1, the circular-variance conflict index (2) identifies sustained opposing-flow windows, and CONFLICT is the dominant cause in 66–71% of frames—so the REDIRECT pathway is *triggered on real data*, in contrast to prior density-only evaluations that never exercised it. Fig. 1 shows the conflict index over time with the REDIRECT frames marked. The ripple-effect *logic* is verified on a controlled two-zone case: with an upstream CONFLICT zone, a downstream score of 0.55 or above downgrades REDIRECT to HOLD, while 0.45 retains REDIRECT. On this particular sequence the scene is sparse (maximum score ≈ 0.50), so no downstream zone ever crosses the 0.50 danger threshold and no downgrade is warranted; we therefore demonstrate the downgrade on the controlled case rather than overclaim it on the real sequence (Sec. V).

E. Multi-Camera Fusion (Scenario D)

Scenario D is a genuine multi-camera evaluation over three overlapping, calibrated real views of the same PETS2009 S2.L1 junction. Per-frame counts disagree substantially across views (mean cross-view spread 4.18 persons, maximum 11.29), which robust median fusion reconciles into a single stable estimate (Fig. 2). We describe this as real multi-view fusion at the scene's actual (low) density; it is not a crush scenario, and no physical venue deployment is implied.

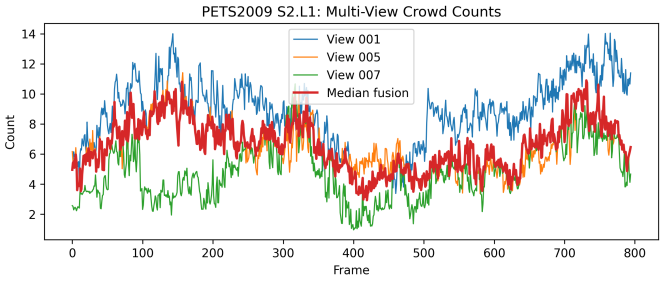


Fig. 2. Per-view and robust-median-fused counts over PETS2009 S2.L1. Views disagree by 4.18 persons on average (up to 11.29); fusion reconciles them.

TABLE VI
SENSITIVITY OF DANGER CLASSIFICATION TO SCORING PARAMETERS

Parameter	**%DANGER range**	**Swing (pts)**	**Effect**
w_1	49–59	10	moderate
w_2	55–55	0	none
w_3	55–55	0	none
γ	42–75	34	strong
λ	30–78	48	strong

F. Parameter Sensitivity

Table VI reports how the DANGER classification responds as each of the five scoring parameters is varied over its range with the others fixed. The classification is *robust to the linear weights* (w_1 moves the DANGER rate by ≤ 10 points; w_2, w_3 by 0) and *sensitive only to the exponential parameters* γ and λ (34 and 48 points). This isolates γ, λ as the parameters requiring careful selection and provides the requested sensitivity analysis over $\{w_1, w_2, w_3, \gamma, \lambda\}$.

G. Throughput

Table VII reports *measured* end-to-end throughput on the Tesla T4: 4.0 frames/s (0.250 s/frame). Counting and motion dominate the budget; the proposed scoring, attribution, gate-mapping, and orchestration stages together run at over $25{,}000$ frames/s and are negligible. At this rate the system provides a 1–2 Hz risk-update cadence, which is appropriate because crowd density evolves over seconds; we do not claim frame-level real-time operation.

V. DISCUSSION AND CONCLUSION

What the results establish. The non-linear score detects crush states that a matched linear model cannot (Table V); the counting front end is validated against ground truth at near-state-of-the-art accuracy on dense imagery (Table IV); the REDIRECT pathway—never exercised in prior density-only evaluations—is triggered on real bidirectional-flow data (Fig. 1); multi-view fusion reconciles genuine cross-camera disagreement (Fig. 2); and the full pipeline runs at a measured, honest 4 frames/s on commodity hardware (Table VII).

TABLE VII
MEASURED END-TO-END THROUGHPUT (NVIDIA TESLA T4)

Stage	**s/frame**	**FPS**
Counting	0.096	10.5
Motion (optical flow)	0.155	6.5
Scoring / attribution / gates / GRS	< 0.001	$> 25{,}000$
End-to-end	**0.250**	**4.0**

Limitations. We state these plainly. (i) PETS2009 S2.L1 is sparse ($\lesssim 4$ persons/m^2, below the crush regime), so it exercises the conflict, motion, and multi-view pathways but not crush-level density, which is why the density pathway is evaluated on dense ShanghaiTech imagery. (ii) The ripple *downgrade* is demonstrated on a controlled two-zone case; the real sparse sequence presents no simultaneously-crowded downstream zone, so no downgrade is warranted on it. (iii) The density pathway is evaluated on single frames ($S_{\text{norm}}{=}C_{\text{norm}}{=}0$), isolating density but without temporal dynamics, and ShanghaiTech lacks metric-area calibration, so capacity is swept rather than fixed. (iv) The reduction coefficients of Table III are assumptions; the projected reduction figures from (8) are therefore simulation estimates conditional on those values, not empirical measurements, and deriving ξ from pedestrian-flow theory is future work. (v) Counters exhibit a cross-domain undercount bias, and throughput supports a 1–2 Hz cadence rather than frame-level real-time. (vi) A dedicated running-crowd sequence was unavailable, so the speed-driven CLOSE pathway is exercised through the scorer; on the real walking sequence the calibrated speed correctly does not raise a speed emergency.

Conclusion. CA-CRS+ integrates cause attribution, gate-specific action with ripple-effect prevention, resource-aware alerting, and genuine multi-view fusion into a single RGB-only, commodity-hardware crowd-safety pipeline, with each component validated on public benchmarks and every quantitative claim either measured against ground truth or explicitly labeled a conditional projection. Future work includes deriving the intervention coefficients from pedestrian-flow theory, evaluating on crush-density video with metric calibration, and physical multi-camera venue deployment with automated gate actuation.

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